Chapter 04 – Cognitive Aspects

Summary

1. Introduction to Cognition in HCI

- **Definition**: Cognition encompasses mental processes like attention, memory, perception, learning, and decision-making.
- **Relevance to HCI**: Understanding cognition helps designers create interfaces aligned with human mental capabilities and limitations.

2. Key Cognitive Processes

A. Attention

- **Selective Focus**: Users concentrate on relevant stimuli while filtering distractions.
  - *Design Implications*: Use color, spacing, and animation to guide attention (e.g., grouping hotel prices in tables vs. cluttered lists).
- **Multitasking**: Heavy multitaskers struggle with focus; interfaces should minimize irrelevant distractions (e.g., FocusMe app blocks notifications).

B. Perception

- **Visual Clarity**: Ensure legible text, distinguishable icons, and effective contrast (e.g., avoid yellow text on white).
  - *Example*: Borders/white space improve information search compared to color contrast alone.

C. Memory

- **Recall vs. Recognition**:
  - *Recall*: Harder (e.g., command-line interfaces).
  - *Recognition*: Easier (e.g., menus, icons, browser history).
- **Design Strategies**:
  - Reduce cognitive load (e.g., password managers like LastPass).
  - Support recognition (e.g., tagging files with metadata).

D. Learning

- **Incidental vs. Intentional**:
  - People prefer "learning by doing" over manuals (e.g., exploratory UI tutorials).
- **Tools**: Multimedia, VR, and dynamic feedback enhance learning.

E. Language Processing

- **Modes**: Reading, speaking, listening.
  - *Design Tips*: Keep voice commands short; amplify text for dyslexic users.

3. Cognitive Frameworks

A. Mental Models

- **Definition**: Users' internal understanding of how systems work (shallow vs. deep).
  - *Example*: Erroneous thermostat models (e.g., turning dial higher heats room faster).
- **Improving Models**: Clear affordances, tutorials, and transparent feedback (e.g., elevator button feedback).

B. Distributed Cognition

- **Concept**: Cognition extends to tools/environment (e.g., using maps for navigation).
- **Application**: Design systems that offload memory (e.g., Autographer camera for dementia patients).

4. Design Implications

- **Attention**: Structure interfaces to minimize distractions (e.g., hospital dashboards).
- **Memory**:
  - Prioritize recognition (e.g., graphical over command-line interfaces).
  - Use metadata (tags, timestamps) for file retrieval.

- **Learning**: Interactive tutorials over static manuals.
- **Error Prevention**: Constrain actions to prevent mistakes (e.g., undo functions).

5. Challenges & Dilemmas

- **Digital Overload**: Apps may reduce decision-making autonomy (e.g., over-reliance on recommendation algorithms).
- **Security vs. Usability**: Multifactor authentication increases security but burdens memory (solved via biometrics).
- **Forgetting**: Tools for "digital closure" (e.g., abstracting ex-partner photos).

6. Key Takeaways

1. **User-Centered Design**: Align interfaces with cognitive strengths (recognition > recall).
2. **Reduce Cognitive Load**: Simplify tasks, provide feedback, and support exploration.
3. **Leverage Frameworks**: Mental models and distributed cognition inform intuitive designs.
4. **Ethical Considerations**: Balance automation with user autonomy (e.g., app dependency).

Quote: *"People prefer to learn by doing."* (Slide 46)

Example:

- **Poor Design**: Cluttered hotel price list (5.5 sec search time).
- **Improved Design**: Grouped table (3.2 sec search time).

This summary highlights how cognitive principles shape interaction design, ensuring usability, efficiency, and user satisfaction.